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Lab2



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Objective

The primary objective of this experiment is to design, implement, and analyze the performance of feedback control strategies for a liquid level control system. Specifically, the experiment aims to:

1. **Evaluate Proportional (P) Control:** To characterize the transient and steady-state behavior of the system under Proportional-only control, identifying its limitations regarding steady-state error (offset).
2. **Evaluate Proportional-Integral (PI) Control:** To demonstrate the effect of adding an Integral term to the control law, specifically its ability to eliminate steady-state error.
3. **Performance Comparison:** To compare the two control strategies based on critical performance metrics, including rise time, maximum overshoot (M_p), settling time, and steady-state accuracy, highlighting the trade-off between system stability and tracking accuracy.

Theoretical Background

2.2 Proportional (P) Control The simplest form of feedback control is Proportional control, where the control signal is directly proportional to the error:

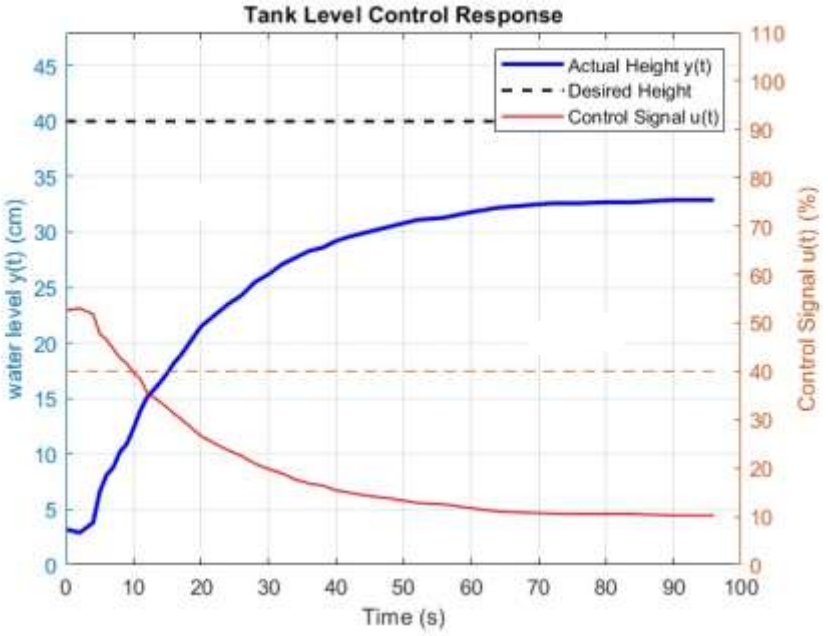
$$u(t) = K_p e(t)$$

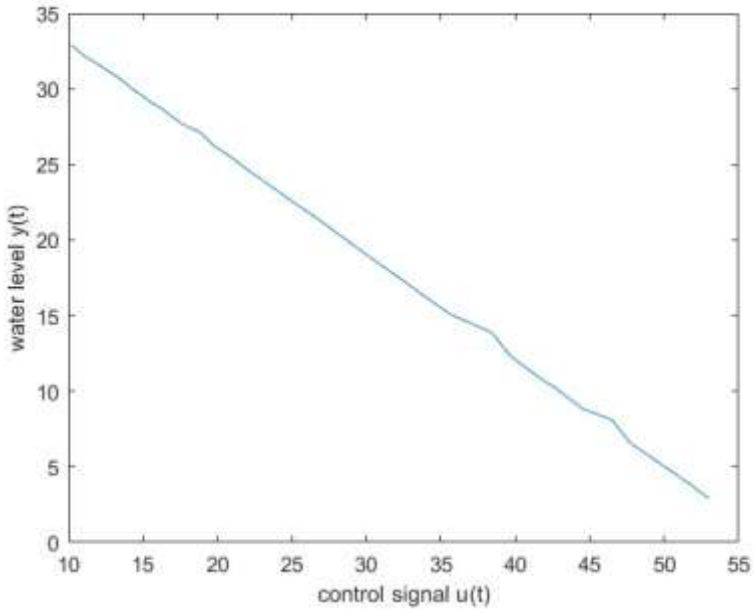
2.3 Proportional-Integral (PI) Control To eliminate the steady-state error, an Integral term is added to the controller. The control law becomes:

$$u(t) = K_p e(t) + K_i \int_0^t e(\tau) d\tau$$

- α (Attenuation Constant): Causes exponential decay of the field amplitude ($e^{-\alpha z}$).
- δ_s (Skin Depth): The distance the wave travels before its amplitude decays to $1/e$ of its initial value, defined as $\delta_s = 1/\alpha$

Proportional Controller





1.Experimental Data Analysis & Controller Characterization

The collected experimental data reveals a distinct linear relationship between the control effort, $u(t)$, and the system error, $e(t)$, defined as $e(t) = r(t) - y(t)$. A linear regression analysis performed on the control signal versus the error yielded an R^2value of approximately 0.9997, indicating a near-perfect linear correlation.

From this analysis, the Proportional Gain (K_p) was determined to be the slope of the regression line:

$$K_p \approx 1.43$$

The control law governing the system behavior was confirmed to be:

$u(t) = 1.43 \times (40 - y(t))$This confirms that the controller is a purely Proportional (P) Controller with a gain of 1.43.

2. Steady-State Performance Analysis

The system response stabilizes at a final output value of $y_{ss} = 32.9$. Given the desired setpoint of $r = 40$, the system exhibits a significant steady-state error (offset):

$$e_{ss} = 40 - 32.9 = 7.1$$

This offset of 7.1 units (representing a 17.75% error) is a fundamental characteristic of a Proportional Controller acting on a Type 0 system (a system without an integral term). For the actuator to maintain the tank level at 32.9, it requires a non-zero control signal ($u \approx 10.2$). Since $u(t) = K_p \cdot e(t)$, a non-zero control signal can only be generated if there is a non-zero error. Therefore, the system reaches equilibrium at a point where the error is just large enough to sustain the necessary control effort.

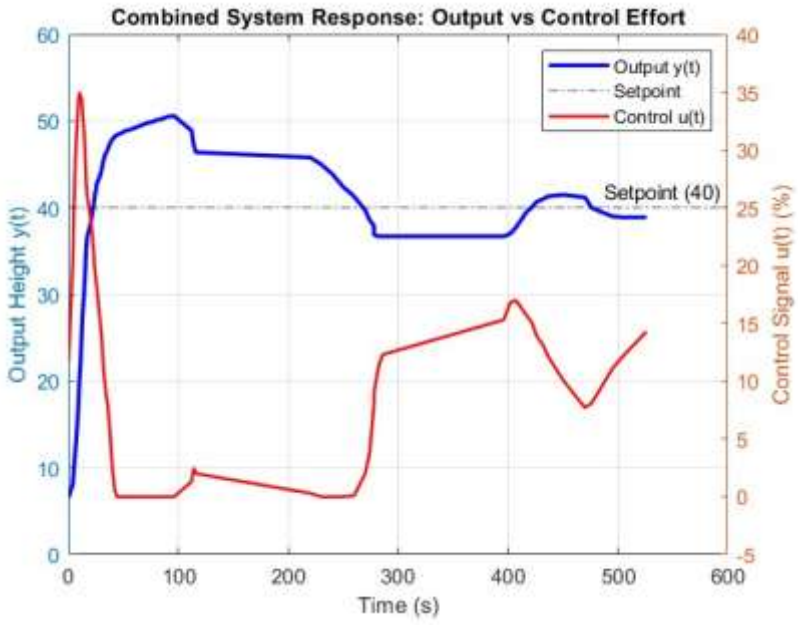
3. Transient Response

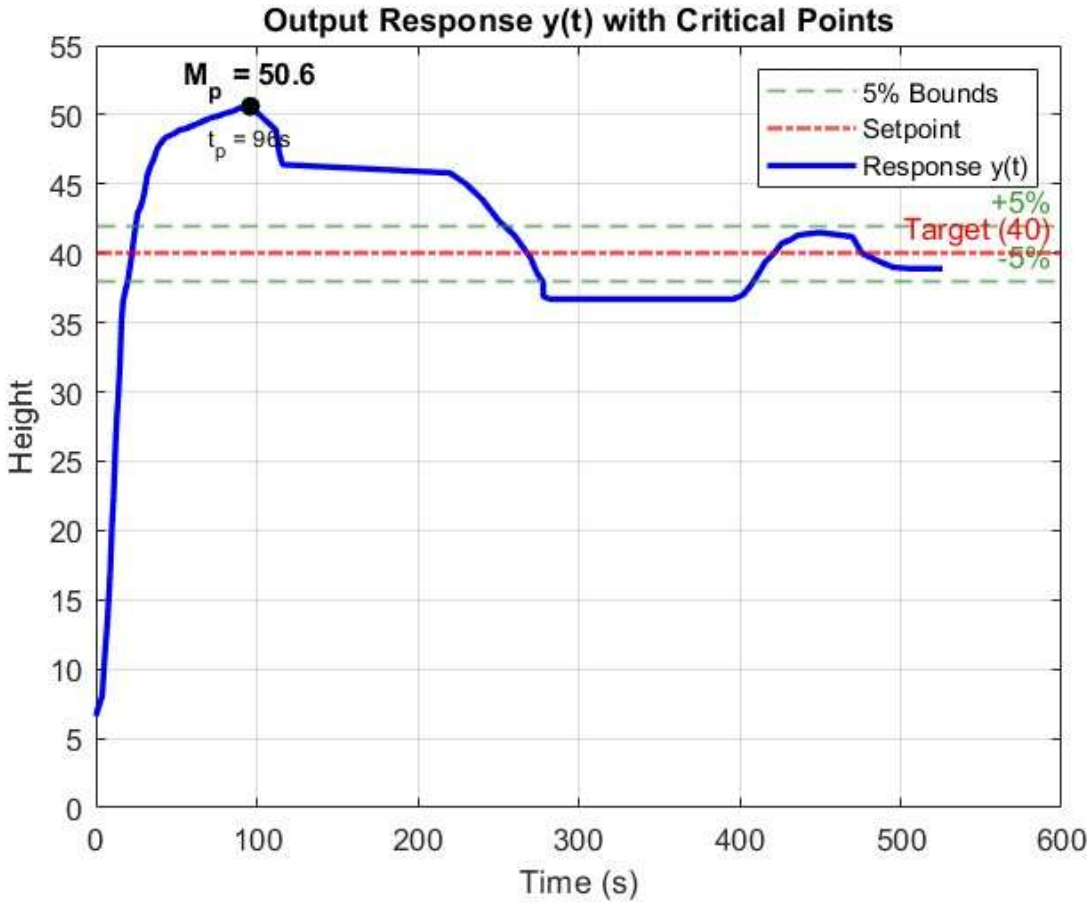
Analysis The transient response of the system is overdamped. The output approaches the steady-state value monotonically without any overshoot or oscillation. This indicates that the selected gain ($K_p = 1.43$) provides a stable response but is insufficient to drive the error to zero or to achieve a rapid rise time. The system takes approximately 90 seconds to fully stabilize, suggesting a relatively slow response time.

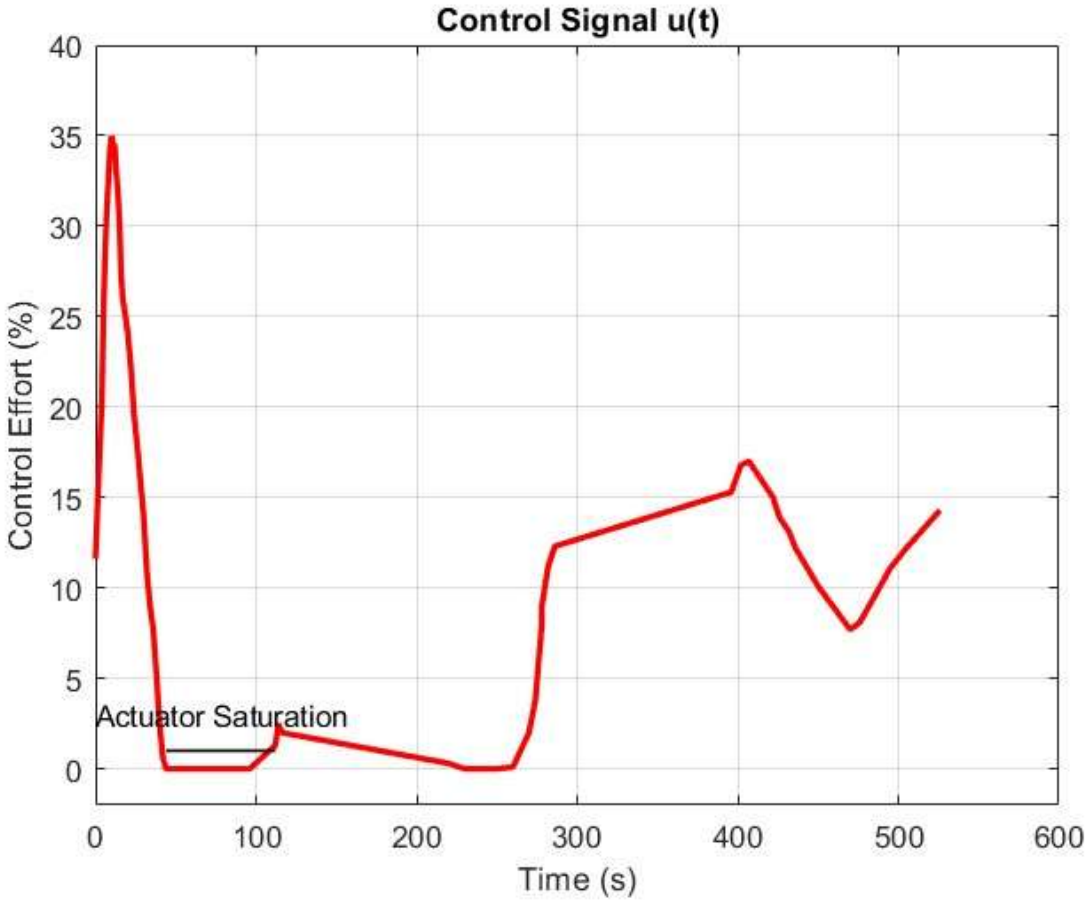
4. Conclusion

The experiment successfully verified the operation of a Proportional Controller. While the controller ensures a stable and non-oscillatory response, it fails to meet the design requirement of zero steady-state error, resulting in a permanent offset of 7.1 units. To eliminate this error while maintaining stability, the control strategy should be upgraded to a Proportional-Integral (PI) Controller, where the integral term would accumulate the residual error and drive the steady-state offset to zero.

PI Controller Analysis




Output Response y(t) with Critical Points
M_p = 50.6
t_p = 96s
5% Bounds
Setpoint
Response y(t)
+5%
Target (40)
-5%
Height
Time (s)
0
5
10
15
20
25
30
35
40
45
50
55
0
100
200
300
400
500
600


Control Signal u(t)
Actuator Saturation
Control Effort (%)
Time (s)
0
5
10
15
20
25
30
35
40
0
100
200
300
400
500
600

1.Experimental Data Analysis & Controller Characterization

The experimental data for the Proportional-Integral (PI) controller demonstrates significantly different behavior compared to the Proportional-only case. While the P-controller exhibited a smooth, overdamped response with a permanent offset, the PI controller data reveals a dynamic, underdamped response characterized by overshoot and oscillation.

Setpoint Tracking: The system attempts to track the setpoint ($r = 40$) but initially exceeds it, reaching a peak value of $y_{max} \approx 50.6$. This confirms the presence of the Integral term, which continues to drive the control signal even as the error decreases, eventually causing the system to overshoot the target.

Control Signal Dynamics: The control signal $u(t)$varies dynamically, notably saturating at $u = 0$for extended periods (e.g., $t = 44$to $t = 112$). This saturation occurs when the output exceeds the setpoint ($y > 40$), causing the error to become negative. The controller attempts to reduce the output by cutting the control effort to zero, but the system's momentum (and likely integrator windup) keeps the output high for a significant duration.

2.Steady-State Performance Analysis

Unlike the P-controller, which stalled at $y = 32.9$, the PI controller successfully drives the system output past the steady-state limitation.

1. 1.Error Reduction: At the end of the data log ($t = 526$ s), the output is $y \approx 38.9$, which is much closer to the target of 40 than the P-controller achieved.
2. Convergence: The system is slowly converging towards the setpoint. The steady-state error is being eliminated, albeit slowly, which validates the theoretical purpose of the Integral action: to remove steady-state offset.

3.Transient Response Analysis

The transient response is underdamped and oscillatory

1. Overshoot: The system exhibits a massive overshoot of approximately 10.6 units (26.5% above the setpoint).
2. Settling Time: The system has not fully settled even after 500+ seconds. It oscillates (dropping to 36.7 before rising back to 38.9), indicating that the PI controller gains are likely not optimal.
3. Integrator Windup: The extended overshoot and saturation of $u(t)$strongly suggest integrator windup. During the rising phase, the error is positive for a long time, causing the integral term to accumulate a large value. When the system reaches the setpoint, this accumulated "momentum" keeps the control signal high, pushing the system well past the target before the error can become negative enough to unwind the integrator.

Conclusion

The PI controller successfully addresses the steady-state error problem identified with the P-controller, allowing the system to reach and even exceed the target height. However, this comes at the cost of stability and settling time. The current tuning results in excessive overshoot and oscillation. To improve performance, the controller requires tuning: likely reducing the Integral Gain (K_i) to reduce overshoot/oscillation, or implementing anti-windup strategies to handle the actuator saturation.

Matlab Code Used to Plot

```
clc;clear;close all;


time=[0,2,4,5,6,7,8,9,10,11,12,13,14,15,16,17,20,22,24,26,28,30,32,34,36,38,40,42,44,48,52,56,60,64,68,72,76,80,84,90,96];

u=[52.6,53,51.8,47.7,46.5,44.6,42.7,41.6,39.7,38.4,35.6,34.6,33.6,32.5,31.2,30.2,26.6,25.1,23.7,22.5,20.9,19.8,18.9,17.6,16.8,16.4,15.4,14.9,14.4,13.7,12.8,12.5,11.7,11.02
,10.8,10.6,10.5,10.5,10.5,10.2,10.2];

y=[3.2,2.9,3.8,6.6,8.1,8.8,10.2,10.9,12.3,13.9,15.1,15.8,16.5,17.3,18.2,18.9,21.5,22.5,23.5,24.3,25.5,26.2,27.1,27.7,28.3,28.6,29.2,29.6,29.9,30.5,31.1,31.3,31.8,32.2,32.4,3
2.6,32.6,32.7,32.7,32.9,32.9];

desired_height=(time.*0)+40;


% 1. Setup the Figure

figure;

title('Tank Level Control Response');

xlabel('Time (s)');

grid on;


% 2. Left Axis: Plot the Heights (Desired vs Actual)

yyaxis left

plot(time, y, 'b-', 'LineWidth', 2); hold on;

plot(time, desired_height, 'k--', 'LineWidth', 1.5);

ylabel('water level y(t) (cm)');

ylim([0 max(desired_height)*1.2]); % Give it some headroom


% 3. Right Axis: Plot the Control Signal (%)

yyaxis right

plot(time, u, 'r-', 'LineWidth', 1);

ylabel('Control Signal u(t) (%)');

ylim([0 110]); % Fixed range for percentage (0-110%)

hold on;

plot(time,desired_height);

% 4. Legend

legend('Actual Height y(t)', 'Desired Height', 'Control Signal u(t)');


figure;

plot(u,y);

xlabel("control signal u(t)")

ylabel("water level y(t)")
```

```
time2=[0,2,4,5,6,7,8,9,10,11,12,13,14,15,16,17,20,22,24,26,28,30,32,34,36,38,40,42,44,48,52,56,60,64,68,72,76,80,84,90,96,112,114,116,220,230,240,250,260,
270,274,278,278,282,286,396,402,407,413,416,422,426,432,436,442,450,470,476,485,495,505,516,526];

u2=
[11.6,16.1,20.4,25.3,28.3,30.8,32.4,34.2,35,34.5,34.5,32.9,31.8,30.2,27.3,26,24.2,22.3,19.6,17.9,15.9,14,10.9,9,7.7,5.2,2.4,.6,0,0,0,0,0,0,0,0,0,0,0,0,0,1.3,2.5,2,0.
3,0,0,0,.1,2,3.8,8,9,11.2,12.3,15.3,16.8,17,16.2,15.8,15,13.9,13.1,12.2,11.3,10.1,7.7,8.1,9.5,11.1,12.2,13.3,14.3];

y2=
[6.6,7.4,8.1,10.1,11.6,13.1,15,17.2,20.3,22.3,25.5,28,29.8,32,35.2,36.6,38.1,39.6,41.4,42.9,43.4,44.3,45.6,46.3,46.8,47.5,47.9,48.2,48.4,48.6,48.9,49,49.2,49.4,4
9.6,49.8,49.9,50.1,50.2,50.5,50.6,48.9,47.2,46.4,45.8,45,43.9,42.4,41.3,39.7,38.6,37.9,36.9,36.7,36.7,36.7,37,37.7,38.8,39.4,40.1,40.7,41,41.3,41.4,41.5,41.2,40,
39.5,39,38.9,38.9,38.9];

% --- 2. Define Critical Values ---

setpoint = 40;

upper_bound = setpoint * 1.05; % 42

lower_bound = setpoint * 0.95; % 38

[max_y, idx_max] = max(y2);

time_at_max = time2(idx_max);



% =========================================================

% FIGURE 1: Combined Plot (Dual Axis)

% =========================================================

figure('Name', 'Combined Response', 'Color', 'w');

grid on; box on;

yyaxis left

plot(time2, y2, 'b-', 'LineWidth', 2);

ylabel('Output Height y(t)');

yline(setpoint, 'k-.', 'Setpoint (40)'); % Simple setpoint reference

ylim([0 60]);

yyaxis right

plot(time2, u2, 'r-', 'LineWidth', 1.5);

ylabel('Control Signal u(t) (%)');

ylim([-5 40]);

title('Combined System Response: Output vs Control Effort');

xlabel('Time (s)');

legend('Output y(t)', 'Setpoint', 'Control u(t)', 'Location', 'northeast');

% 2. Plot Setpoint (Red)

yline(setpoint, 'r-.', 'LineWidth', 1.5, 'Label', 'Target (40)', 'LabelHorizontalAlignment', 'right');
```

```
% ========================================================
% FIGURE 2: Output y(t) ONLY (With Analysis Labels)
% ========================================================
figure('Name', 'Output Analysis', 'Color', 'w');
hold on; grid on; box on;
% 1. Plot 5% Error Bands (Subtle Dark Green)
% Using a custom dark green RGB [0.2 0.6 0.2] and thinner line
yline(upper_bound, 'Color', [0.2 0.6 0.2], 'LineStyle', '--', 'LineWidth', 1, 'Label', '+5%');
yline(lower_bound, 'Color', [0.2 0.6 0.2], 'LineStyle', '--', 'LineWidth', 1, 'Label', '-5%');
% 3. Plot Actual Response (Blue)
plot(time2, y2, 'b-', 'LineWidth', 2);
% 4. Mark Maximum Overshoot (Mp)
plot(time_at_max, max_y, 'ko', 'MarkerFaceColor', 'k');
text(time_at_max, max_y + 2, ['M_p = ' num2str(max_y)], 'HorizontalAlignment', 'center', 'FontWeight', 'bold');
text(time_at_max, max_y - 2.5, ['t_p = ' num2str(time_at_max) 's'], 'HorizontalAlignment', 'center', 'FontSize', 8);
title('Output Response y(t) with Critical Points');
xlabel('Time (s)');
ylabel('Height');
ylim([0 55]);
legend('5% Bounds', '', 'Setpoint', 'Response y(t)');
% ========================================================
% FIGURE 3: Control Signal u(t) ONLY
% ========================================================
figure('Name', 'Control Effort', 'Color', 'w');
hold on; grid on; box on;
% 1. Plot Control Signal (Red)
plot(time2, u2, 'r-', 'LineWidth', 2);
% 2. Mark Saturation (u=0)
% Find the saturation region manually for labeling
sat_start = 44;
sat_end = 112;
% Draw a bracket or line indicating this region
line([sat_start sat_end], [1 1], 'Color', 'k', 'LineWidth', 1);
text((sat_start+sat_end)/2, 3, 'Actuator Saturation', 'HorizontalAlignment', 'center');
title('Control Signal u(t)');
xlabel('Time (s)');
ylabel('Control Effort (%)');
ylim([-2 40]);
```